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ABSTRACT

Whether organizational performance shapes government resource allocations or not has long been a contentious issue, especially since the 1990s when the New Public Management movement vitalized performance-based budgeting across countries. To contribute to this ongoing debate, this study examines if different levels of organizational performance achievement affect annual budget changes, focusing on Washington State's public programs from 2006 to 2015. This study finds a significant relationship between program performance and budget changes, particularly during fiscally hard times. The findings also suggest the reward mechanism associated with performance-based resource allocation, indicating that higher program performance may lead to budget increases.

KEYWORDS

budget allocation; fiscal condition; organizational performance; performance-based budgeting; Washington State

Introduction

The use of performance measurement and information has been one of the most significant governmental reforms over the last several decades. Performance measurement systems are expected to track organizational goals, targets, and achievements while offering information that helps improve the quality of government decision-making and service delivery (Julnes & de Holzer, 2001; Moynihan, 2009). Public officials can also use performance measurement for communication purposes, thereby better elucidating their decisions and actions to citizens (Korac et al., 2020). Studies have consistently documented that using performance measurement and information directly leads to positive outcomes, such as robust strategic planning, enhanced government effectiveness, greater accountability, and improved citizen trust (Hvidman & Andersen, 2014; Wang, 2000). Conditions that facilitate performance information use, such as stakeholder involvement and leadership support, have also garnered attention from academics and practitioners (Kroll, 2015).

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Performance information use is now ubiquitous in almost all aspects of government operation, and budgeting is not an exception. While the history of using performance information for government budgetary purposes dates back to the 1940s (Ho, 2018), it was the New Public Management movement in the 1990s that vitalized performance-based budgeting (PBB) initiatives in the U.S. and across other countries (E. Y. Lu et al., 2015). In the spirit of PBB, past studies have illuminated various changes that performance information has brought into the government budget process, such as informed budgetary communications and coordination and enhanced management and accountability (Hou et al., 2011; J. Melkers & Willoughby, 2005; Moynihan, 2006). What is still inconclusive—and even remains controversial—is whether performance information alters government budget allocations, indicating potential dynamics involved in the performance-resource allocation link (see also Robinson & Brumby, 2005). This question about the allocative efficiency of PBB is challenging, yet essential to understand the precise role of performance information use in budgeting and, more broadly, government operation.

This study contributes to the crucial debate by examining whether, when, and how different levels of performance achievement, captured by performance information systems, affect annual program budget changes. Upon demonstrating the association between performance and budget changes, this study considers a set of competing hypotheses from the organizational rationality perspective. First, public officials may connect performance achievement systematically to budget allocations when facing fiscal stress or affluence. In either case, they may have distinct motivations in whether or not to use performance information for budget allocation purposes, as their desires for rational decision-making could vary. Second, performance achievement can lead to increased or decreased program budgets. When tying performance to resource allocations, a reward or supplement mechanism can be a rational option for public officials. Washington's public programs from 2006 to 2015 offer a suitable setting for this study given the state government's long history with PBB since 1994 and its emphasis on objective performance measurement using clear benchmarks (Barrett & Greene, 2008; Hou et al., 2011; J. Melkers & Willoughby, 1998).

This article proceeds as follows: the following section reviews the performance management and PBB literature and proposes research hypotheses. The next section discusses methodology issues, including variables, measures, and estimation strategies. A discussion of empirical findings follows. The final section suggests implications and conclusions.

Literature review

Performance and budget allocation: competing ideas

When the Government Performance and Results Act was enacted in 1993, President Clinton (1993) described it as a means of achieving "greater efficiency and effectiveness and less unnecessary cost" by charting "a course for every endeavor that we take the people's money for" (p. 1540). When California legislators agreed on the Performance and Results Act in the same year, they wanted to reform their existing budget process where inefficient programs "may be rewarded" because it made "allocations … based on the prior year's budget" (Hill, 1993). These examples illustrate lawmakers' early expectations of performance-based reforms to improve managerial and operational efficiency and minimize wasteful spending through performance-based resource allocation (Crain & O'Roark, 2004; Lee & Wang, 2009; Moynihan, 2003; Robinson & Brumby, 2005; Shah & Shen, 2007).[1]

Despite these promises, the question as to whether organizational performance achievement can be tied to actual budget decisions and generates a variation in government resource allocations has been met with mixed answers. Competing ideas exist due to different views on organizational rationality (Robinson & Brumby, 2005).[2] One group of studies has advocated a plausible link between organizational performance achievement and subsequent resource allocations in the budget process. Decision-makers and other government officials, rationally motivated to achieve their goals, may find performance information valuable in seeking more efficient decision-making and program implementation. Legislatures, for example, may want to know agencies' performance achievements to mitigate principal-agent problems; executive agencies can improve their cost structures by looking at their performance results (Douglas et al., 2019; Miller et al., 2007). Besides this instrumental value, public officials may want to link performance achievement to resource allocations to justify their actions and reduce public cynicism (Berman, 1997; Korac et al., 2020; Miller et al., 2007). This may lower the political costs involved in their budgetary decisions, which can help realize their rational interests.

Previous studies have offered evidence consistent with this rationality idea. Jordan and Hackbart (1999) surveyed state executive budget officers. According to the results, 29 and 33 out of 45 respondents (64.4 and 71.7%, respectively) agreed that "the achievement of performance standards" (p. 75) affected executive budget recommendations and the level of performance achievement in the current year affected "funding for the next fiscal year" (p. 77). Ho (2006) reported a similar result at the local level: 59% out of 244 Midwestern mayors agreed that performance accomplishment

information helped "elected officials determine the funding level of a department" (p. 224). While examining PBB's impacts on overall government finances such as lower total expenditures, deficits, or debt levels, some other studies have also considered performance-based allocation decisions as one reason for PBB's significant fiscal influences (Crain & O'Roark, 2004; Klase & Dougherty, 2008; Lee & Wang, 2009).

However, the other group of studies has suggested the opposite idea: linking performance to budget allocations is not an easy task in reality. Their primary focus is on the political, value-driven, and incremental aspects of government budgeting (J. E. Melkers et al., 2002; Shah & Shen, 2007; Willoughby & Melkers, 2000). The government budget process is inherently political, as it usually involves varying actors with complicated interests and relationships (Wildavsky, 1975). It is, therefore, seldom possible to find sufficient room for rational decision-making and informed allocations. As a more practical reason, researchers have also pointed out the ambiguity of performance measurement (J. E. Melkers & Willoughby, 2006; Moynihan, 2006; Murray & Alev, 2011). This potential caveat can cause significant value conflicts and distrust in performance information (Moynihan, 2003), making connecting performance achievement to resource allocations more complicated and less feasible (Robinson & Brumby, 2005; Smith, 1995). Moynihan (2006) asserted that PBB could help the budget process by offering information, but "funding judgments … may cumulatively not demonstrate a strong influence of performance information and may even appear to be random" (p. 167).

Based on a survey of state budget officers, J. E. Melkers et al. (2002) showed that only 7.5% of 60 respondents agreed with the effectiveness of performance measures in changing budget appropriations. Moreover, only 26.3% of total respondents said that performance measures are "important" or "very important" in making budget appropriation decisions. Lu (2007) also found that state officials in Georgia used performance data to a lesser extent as the budget process moved from preparation to appropriation steps. The author concluded that "the uses of performance information by agencies focused on managing and improving performance with a very modest realignment of resource allocation decisions" (p. 12). From a survey of 695 local officials, Poister and Streib (1999) found that performance measures only have moderate or slight impacts on their budget allocation process. Similarly, Reck (2001) surveyed city officials and found that they tended not to connect performance to their allocation decisions. Ryu (2013) also documented no allocative efficiency of PBB at the federal level through a case study of the U.S. Department of Commerce.

Fiscal condition and performance-based budget allocation

The competing ideas discussed above suggest potential dynamics involved in the association between performance achievement and budget allocations. The link, in other words, could be conditional based on different environmental, organizational, and individual attributes (Grizzle & Pettijohn, 2002; Kelly & Riverbark, 2002; Y. Lu, 2011; Ryu et al., 2007, 2008; Thompson, 1987; Wang, 2000). One potential moderating factor that has recently attracted scholarly and practical attention but warrants more research is government fiscal condition. The core question is if fiscal affluence or stress increases the need for rational decision-making in the budget process and helps public officials' desires for informed choices.

Public officials may tie performance accomplishment more closely to budget allocations when fiscally affluent and stable. Rational decision-making and implementation using performance information is costly, as it takes time and resources (Frølich, 2011; McNab & Melese, 2003; J. E. Melkers & Willoughby, 2001). Abundant resources available in a fiscally affluent period may help public officials seek more rationality in the budget process by allocating resources for the use of performance information (Askim et al., 2007; Pfiffner et al., 2021). Such resource availability can also make performance information use more cost-beneficial to public officials, as fiscal affluence may enable them to invest or receive more resources for their programs (Hou et al., 2011; Johansson & Siverbo, 2009). From this resource availability standpoint, performance achievement is likely to be decoupled from budget allocations when resources are scarce (Kroll, 2013). Public officials may have less bandwidth for "rational-analytical considerations" (Raudla & Bur, 2022, p. 2), lacking time and resources available for performance-based allocations in the budget process.

On the other hand, the performance-allocation link can be more desirable when public officials face fiscal stress or austerity. This idea builds on the behavioral observation that problem-solving can offer a clear and even stronger motivation to use information in an organization (Spink & Cole, 2006). Fiscal stress that results in acute resources is a challenging issue; public officials may have a solid motivation to connect performance to budget allocations to solve the problem (Bjørnholt et al., 2016). Organizational performance can inform public officials how to reduce or cut their budgets (Pollitt, 2010; van der Voet, 2019). It can also help public officials find strategic directions and priorities to maintain the quality and quantity of public services (Cayer, 1986). Fiscal affluence may undermine public officials' motivation to link performance to resource allocations through this lens. When fiscally stable, they have space and resources to simply add up to the existing budget trend (i.e., incrementalism) or make innovative policy changes that they would not have tried without bountiful

resources (Chernick & Reschovsky, 2017; Ibrahim & Proctor, 1992; Le Maux & Zhang, 2012).

Studies have reported mixed empirical evidence. Hou et al. (2011), through a case study of 11 state governments, offered that "PBB is used more by the states during strong economic times as opposed to during economic downturns" (p. 370). The authors observed that agencies, for example, used PBB as "a bargaining tool" (p. 376) for their budget justifications when fiscally healthy. Askim et al. (2007), Jimenez (2014), Pfiffner et al. (2021), Raudla and Bur (2022), and Raudla and Savi (2015) supported the same idea in different contexts by documenting less or nonsystematic use of performance information during fiscally stressful times. In contrast, Ho's (2011) seminal work focused on Indianapolis' public programs in 2010 when the city was still under the fiscal pressure caused by the Great Recession. His findings suggested that performance measurement at the program level had a significant impact on budget changes amongst programs. Douglas et al. (2019), Good (2011), and Raudla et al. (2017), in varying contexts, also provided evidence consistent with the idea that fiscal stress tends to encourage public officials' interest in organizational performance for budgetary purposes.

Considering these conflicting but equally convincing ideas, this study offers two competing hypotheses as follows:

H1-1: Organizational performance achievement is likely to be associated with annual program budget changes during times of fiscal health.

H1-2: Organizational performance achievement is likely to be associated with annual program budget changes during times of fiscal distress.

Performance-based budget allocation: a reward or supplement mechanism?

Public officials have two viable options when connecting performance achievement to budget allocations (Gilmour & Lewis, 2006a, 2006b; Moynihan, 2003; Nielsen & Baekgaard, 2015). One is a reward-sanction mechanism, meaning that public officials may link higher (lower) performance to an increased (decreased) budget allocation. This mechanism can be a rational strategy because it can help eliminate waste and provide public officials with incentives to perform more effectively. By doing so, this strategy can positively contribute to government efficiency. From this standpoint, Lee and Wang (2009) described that performance-based budget allocation "can be used as a strategy to change spending behaviors by proposing fewer ineffective programs or by eliminating more ineffective existing programs" (p. 63). This approach is also in line with decision-

makers' political interests, as it could be a simple, direct way to react to public backlash against poor government performance and express support for higher government performance (Nielsen & Baekgaard, 2015).

The other option works in reverse: linking higher (lower) performance to a decreased (increased) budget allocation, which can be described as a redress-supplementary mechanism. Under this mechanism, low or under-performance can be attributable to the lack of resources (Coggburn & Schneider, 2003; Walker & Andrews, 2015). Presuming that all agencies and programs serve unique and exclusive functions, certain agencies' or programs' under-performance can be detrimental to attaining the overall goals of a government. Also, according to Nielsen and Baekgaard's (2015) discussion of blame-avoidance theory, public officials "often find it difficult or impossible to disassociate themselves from the performance of key public programs and organizations" (p. 550). Hence, they must propose ideas to improve performance, such as "increased funding" (p. 550). Higher performance, on the other hand, could mean that organizations have enough or slack resources (Park & Jang, 2021). Moving resources from high to low performers could be a rational strategy to better achieve governmental goals.

Only a handful of studies have directly examined the direction of the association between performance achievement and budget allocations (see also Robinson & Brumby, 2005). Gilmour and Lewis (2006a, 2006b), in their examination of Program Assessment Rating Tool (PART) scores and federal program budgets, found a positive impact of program performance on annual budget changes. This evidence supports the reward-sanction idea discussed above. In contrast, Zaltsman's (2009) empirical test on the effect of performance target compliance on agency budget changes suggested a negative result, meaning that higher performance achievement may lead to reduced budgets (i.e., the redress-supplementary mechanism). Nielsen and Baekgaard (2015) added to this conflicting evidence by suggesting that both high and low performance significantly affected politicians' spending attitudes positively. How can politicians meet budget constraints while using a reward-supplementary mechanism? In their examination, politicians tended to cut agency budgets when they found an average level of performance.

These two compelling ideas lead to the following competing hypotheses:

H2-1: Higher organizational performance is likely to be associated with higher annual program budget changes.

H2-2: Lower organizational performance is likely to be associated with higher annual program budget changes.

Methodology

Sample: Washington State's public programs

The unit of analysis is Washington State's public programs. Washington has a relatively long experience with PBB. After adopting PBB in 1994, the state government has continued its efforts to develop and refine objective performance measures and use them for budgeting purposes (Barrett & Greene, 2008; Hou et al., 2011). Hence, the case is expected to minimize caveats related to the potential ambiguity or disagreement of performance measures among the state's public officials. The Governor and the Office of Financial Management (OFM) request executive agencies to annually report their programs' performance targets and actual accomplishment levels over the last few years. The information is integrated into the budget process under the PBB initiative (Washington OFM, 2019). Another significant benefit of the Washington case is that most state programs use a single indicator to measure their performance, which can help researchers measure performance achievement easily and objectively. As one typical case of PBB implementation at the state level, the case is expected to offer novel but reliable observations.

Through Washington's OFM, the program and performance information for about 80 different state executive agencies from 2004 to 2013 was collected (Washington OFM, 2004–2013). The performance dataset covers approximately 5800 programs (580 per year on average). Of those 5800 programs, about 4010 programs have specific performance indicators. Performance achievement is fully measurable for approximately 2000 programs because other programs miss their performance targets or accomplishments. Annual approved budget information (including supplemental changes) was obtained through the Washington Legislative Evaluation and Accountability Program (LEAP) Committee (2006–2015). The budget dataset covers 2006 to 2015, given that a two-year lag between performance and budget data is typically shown in the state's budget information (Washington OFM, 2004–2013). Combining these two major datasets, this study includes about 1750 year-program observations (unbalanced), covering an average of 47 executive agencies each year.

Variable and measurement

The empirical model to test the research hypotheses can be expressed as follows:

$$B_{i,\ t} = \alpha + \beta_1 \cdot P_{i,\ t-2} + \beta_2 \cdot E_{i,\ t-1} + \beta_3 \cdot (P_{i,\ t-2} * E_{i,\ t-1}) + \sum \beta_3 \cdot C_{i,\ t,\ t-1,\ \text{or}\ t-2} + \varepsilon_{i,\ t} \quad (1)$$

where, $B_{i,\ t}$ stands for program i's budget change from $t-1$ to t. $P_{i,\ t-2}$ means program i's performance achievement in year $t-2$ (i.e., lagged two years from the dependent variable). $E_{i,\ t-1}$ is an indicator of economic downturn and fiscal distress. $P_{i,\ t-2} * E_{i,\ t-1}$ is an interaction term between the performance achievement and economic downturn variables. $C_{i,\ t,\ t-1 \text{ or } t-2}$ indicates a vector of program, department, and state-level control variables measured in year t, $t-1$, or $t-2$. $\varepsilon_{i,\ t}$ is a well-behaved error term.

The dependent variable of interest is each program's budget change, measured as a percentage difference from the last year. A positive or negative percentage change means that the program receives more or less budget, respectively, compared to the previous year. As hypothesized, more budgets allocated to specific programs could result from the reward or supplementary mechanism. In contrast, public programs subject to the sanction or redress mechanism may experience less budget allocation than last year.

The key independent variable of interest is expected to capture the different levels of program performance achievement. Inspired by Nielsen and Baekgaard (2015) and Zaltsman (2009), this study measures program performance achievement at three different levels: under, right, and over-performance. While under-performance (coded as zero) means that public programs fail to achieve their target goals, right-performance (coded as one) indicates no difference between targets and actual accomplishment. Over-performance (coded as two), in turn, suggests that public programs achieve higher performance than their target goals. Some of the sample programs have two or more performance indicators. In these cases, each indicator is first measured following the coding scheme above (i.e., zero, one, and two). Then, the sum of those assigned values is divided by the total number of indicators. Programs are under or over-performed when they receive smaller or greater than one; they are right-performed when the average score equals one.[3] The variable is expected to have a positive or negative direction, as hypothesized in the previous section.

The economic downturn indicator is expected to capture the years when the state government was under fiscal stress. According to Washington OFM (2012), the state's fiscal condition started showing the baleful impact of the Great Recession in early 2008, and such impact lasted until 2011. Following this observation, the indicator is coded as one for 2008 to 2011, otherwise zero. The variable is lagged one year, considering the time gap between budget decisions and implementation. It is expected to have a negative impact on program budget changes (Skidmore & Scorsone, 2011).

This indicator interacts with the performance achievement variable, considering the research hypotheses that require separating public officials' use

of performance information during a period of fiscal stress from fiscally affluent times. The interaction term, by doing so, highlights the instances where public officials in fiscally had times ($t-1$) use information about the extent to which public programs achieve their target objectives or not in year $t-2$ to make budget allocations for the next fiscal year (t). As hypothesized, the interaction term is expected to have a positive or negative direction.

The literature suggests two sets of factors that can affect program budget changes: program-level characteristics and broader- or macro-level conditions (see, e.g., Agthe et al., 1996). At the program level, this study considers three different control variables, including the number of performance indicators used to measure each program ($t-2$), each program's budget amount ($1000; $t-1$), and whether general fund resources fund each program or not ($t-1$). How many performance indicators each program uses may affect its annual budget change, especially under PBB, as it determines the amount of information public officials consider during the budget process (Ho, 2011). As baseline budget information, each program's budget amount in the last year may be significant in determining the following year's budget change (Ryu, 2013; Zaltsman, 2009). Whether each program is a general fund program or not is critical given that state-level expenditure limitations often specifically restrict general fund expenditures; Washington falls into this case (Kioko, 2011). Given the mixed findings of previous studies, the first two control variables are expected to have a positive or negative sign in the models. The general fund variable is expected to have a negative direction (data come from Washington LEAP Committee, 2006–2015 and Washington OFM, 2004–2013).

At a broader level, this study considers two departmental variables: the total number of programs in a department in which each program is housed ($t-1$) and the total department budget amount ($1000; $t-1$). The more programs a department has, the higher resource competition—or conflict—may exist amongst those programs (Carrigan, 2018). It could prevent significant increases in program budgets. A variation in the total department budget amount variable may reflect state policy priorities or the relative importance of each department. Higher policy priorities or importance could increase program budgets (Garand, 1985; Jacoby & Schneider, 2009; data come from Washington LEAP Committee, 2006–2015). Three different conventional state-level variables are also considered, including median household income ($1000; $t-1$), unemployment rate ($t-1$), and gubernatorial election years (t). The first two variables capture the socio-economic condition of the state (Klase & Dougherty, 2008; data come from the U.S. Census Bureau, 2005–2014). Given the mixed sign of these variables reported in the literature, expectations for the direction

of these signs are unclear. The election-year variable (data from the National Governors Association, 2022) is expected to have a positive sign. According to the political budget cycle literature, decision-makers can be motivated to spend more in election years (De Haan & Klomp, 2013).[4]

Estimation strategy

The estimation process may be subject to selection bias concerns, mainly due to the structure of the collected data. In response to such concerns, the average budget of all programs in 2015 ($36.2 million) was compared to the average budget amount of the sample programs in the dataset ($26.4 million). The difference was identified but not substantial. In addition, the Heckman selection approach was used to examine these selection bias issues. In the first selection stage, whether programs were included in the dataset or not was regressed on department and year fixed effects. After calculating the inverse mill's ratio, it was included in the second-stage model. The inverse mill's ratio was insignificant across the models, indicating no systematic evidence for selection bias. This study, therefore, employs the OLS estimation. In support of this approach, the results of the major model returned by the Heckman selection estimation, generally consistent with the OLS findings, are reported in Appendix A.

Multilevel regression analysis may be appropriate for this study because the sample programs are grouped into categories due to department-level variables. One necessary condition that makes multilevel regression analysis more feasible and efficient than fixed-effect models is the significance of random effects at the group level. Otherwise, multilevel regression results are biased and inefficient. The likelihood-ratio test was performed after estimating the mixed-effect models to confirm this issue. The results indicated no significant random effects at the group level ($p > .185$); fixed-effect analysis is more appropriate in this case. Instead of controlling for fixed effects that absorb all upper-level variables, this study leaves those department and state-level variables in the models. The expectation is that the variables can account for cross-sectional and time fixed effects to a substantial degree. This strategy can help avoid leaving those fixed effects in a black box. In Appendix A, the supplementary analysis results confirm that the findings are consistent when estimated including department and year fixed effects instead of the department and state-level control variables. Program-level fixed effects are not considered because they overfit the models and produce misleading results.

While considering cross-sectional fixed effects and leaving year lags between performance achievement and budget changes, there could still be endogeneity concerns. Potential sources for these concerns include omitted

variable bias, dynamic endogeneity, and simultaneity; the system generalized method of moments (GMM) estimation could be an effective instrument to deal with endogeneity in this case (Li et al., 2021; Ullah et al., 2018). The models showed the results consistent with the OLS findings when estimated using the system GMM estimation, as shown in Appendix A. This observation confirms no significant endogeneity bias in the models, further suggesting that the OLS estimation can produce efficient results.

Findings

Descriptive statistics

Table 1 presents descriptive statistics. The program budget change variable has a mean of 7.5, indicating that the sample programs experienced an average of 7.5% annual budget increase from 2006 to 2015. Its standard deviation shows a considerable variation among the state's public programs. Program performance achievement has a mean of 1.1, suggesting that Washington's public programs, on average, accomplished right or slightly over-performance from 2004 to 2013. The economic downturn variable shows that less than half of the sample programs fall into the fiscal stress category. The average number of performance indicators per program was 1.3; the average program budget was $14.7 million. More than half of the sample programs were funded by general fund sources, with a mean of 0.7.

Table 1. Descriptive Statistics.

Variable	Measure	Mean	Standard deviation
Dependent variable			
Program budget change	Percentage change from the last year's budget (t)	7.539	54.152
Independent variable			
Program performance achievement	Under (0), right (1), or over (2) performance compared to target ($t-2$)	1.106	0.951
Moderating and control variables			
Economic downturn	Dummies for 2008 to 2011 ($t-1$)	0.451	0.498
Interaction term	Program performance achievement ($t-2$) *economic downturn ($t-1$)	0.501	0.841
Number of performance indicators	Counted number of performance indicators ($t-2$)	1.278	0.731
Program budget amount	Total program budget amount in thousand dollars ($t-1$)	14746	42977
General fund	Dummy (1) for general fund programs, otherwise (0) ($t-1$)	0.655	0.745
Total number of programs	Counted number of programs in a department ($t-1$)	35.890	32.874
Department budget amount	Total department budget amount in thousand dollars ($t-1$)	934,448	1,725,324
Median household income	State median household income in thousand dollars ($t-1$)	56.553	1.713
Unemployment rate	State unemployment rate ($t-1$)	7.036	1.941
Election year	Dummies for 2008 and 2012	0.272	0.445
N	1749		

On average, Washington departments in the dataset had more than 35 programs each year. Their average department budget amount was $934.4 million. The state's average median household income and unemployment rate were $56,553 and 7%, respectively. About 27% of the sample programs were performed in election years.

Regression results

The regression analysis results are reported in Table 2. The main results are available in Model 1, where the independent variable is lagged for two years. Given that Washington executive agencies typically report their program performance over the last three or four years, Models 2 and 3 are additionally estimated and reported in the same table. Each additional model understands the influence of previous performance information by lagging the independent variable for three or four years (e.g., program budget changes in 2015 are regressed on performance in 2012 in Model 2). Logarithmic transformations are used for the program and department budget amount variables.

Each model has two sets of results for comparison—with and without the interaction term. According to the likelihood-ratio test results, the models with the interaction term fit better to the data than their counterparts without the interaction term. The only exception is Model 3, where the interaction term is insignificant. The interpretation below primarily focuses on the interaction models given the likelihood-ratio test results and the scope of the research hypotheses. All models are significant according to the F-statistics. R^2 values range from 0.029 to 0.061. The highest VIF across the models is 4.09, indicating no severe multicollinearity issues.

The key independent variable, program performance achievement, has a positive coefficient in Model 1 but failed to obtain statistical significance. Since the model includes an interaction term, the coefficient means the relationship between the independent and dependent variables when the economic downturn variable is zero (i.e., fiscally affluent). Performance achievement, therefore, has no significant impact on program budget changes during fiscally affluent or stable times. This result rejects H1-1. As discussed, fiscal affluence may diminish public officials' motivation to link performance achievement to budget allocations. They have enough resources for simple incremental budgeting or innovative policy changes and budget investment (Chernick & Reschovsky, 2017; Ibrahim & Proctor, 1992; Le Maux & Zhang, 2012).

The coefficient of the economic downturn variable in this interaction model means the impact of fiscal stress on program budget changes when program performance achievement is zero (i.e., under-performance). The

Table 2. Regression Results.

	Dependent variable: annual program budget change					
Variable	Model 1 (two-year lag)		Model 2 (three-year lag)		Model 3 (four-year lag)	
Program performance achievement	2.985**	0.542	2.771***	0.883	0.692	2.540
	(1.235)	(2.034)	(0.857)	(1.479)	(1.249)	(2.010)
Economic downturn	−3.119	−9.157***	−1.001	−4.992**	−5.681*	−1.651
	(2.309)	(2.796)	(1.995)	(2.503)	(3.030)	(3.221)
Interaction term (program performance achievement*economic downturn)	–	5.463**	–	3.529*	–	−3.318
		(2.718)		(1.933)		(2.388)
Number of performance indicators	1.485	1.549	−0.272	−0.238	0.917	0.917
	(1.206)	(1.198)	(0.879)	(0.878)	(1.481)	(1.490)
Program budget amount (Ln)	−5.919***	−5.924***	−1.546*	−1.615*	−3.136***	−3.110***
	(1.235)	(1.242)	(0.908)	(0.921)	(0.961)	(0.956)
General fund	−3.820	−3.895	−2.763	−2.632	−1.786	−2.023
	(2.835)	(2.847)	(2.086)	(0.205)	(2.472)	(2.475)
Total number of programs	−0.121**	−0.123**	−0.083*	−0.083*	−0.118*	−0.117*
	(0.053)	(0.053)	(0.044)	(0.044)	(0.070)	(0.070)
Department budget amount (Ln)	4.609***	4.681***	2.456***	2.486***	3.823***	3.710***
	(0.947)	(0.957)	(0.771)	(0.772)	(0.902)	(0.896)
Median household income	−1.381***	−1.533***	−0.311	−0.421	−2.265**	−2.144**
	(0.466)	(0.457)	(0.482)	(0.475)	(1.115)	(1.086)
Unemployment rate	−2.885***	−2.924***	−2.351***	−2.354***	−2.397*	−2.353*
	(0.692)	(0.690)	(0.538)	(0.538)	(1.239)	(1.231)
Election year	10.996***	11.014***	−0.495	−0.309	6.037	5.722
	(3.754)	(3.747)	(3.085)	(3.082)	(3.989)	(4.024)
Constant	98.734***	109.687***	22.601	30.881	137.626*	128.457*
	(31.007)	(30.247)	(27.644)	(27.403)	(72.498)	(70.455)
N	1749	1749	1544	1544	1248	1248
F	4.92***	6.03***	5.19***	5.46***	2.78***	2.55***
R^2	0.059	0.061	0.039	0.036	0.029	0.030
Highest VIF	3.39	3.39	3.45	3.77	3.62	4.09
Likelihood-ratio test	–	4.18**	–	3.40*	–	0.22

Notes: Unstandardized coefficients are reported; robust standard errors are in parentheses.
*$p < .10$; **$p < .05$; ***$p < .01$.

variable's statistical significance denotes that fiscal stress may generate a significant budget gap (approximately 9%) among under-performing programs. The interaction term should be of major interest to this study. Its significant coefficient indicates that when fiscally stressed, there is a significant association between performance achievement and program budget changes. Therefore, H1-2 is supported. More specifically, programs may experience a 5% budget increase next year if they achieved or over-achieved their targets last year, supporting the reward-sanction mechanism described in $H_{2\text{-}1}$. Fiscal stress seems to encourage public officials to focus on the problem-solving aspect of performance information use in the budget process, such as eliminating waste and improving efficiency (Cayer, 1986; Pollitt, 2010; van der Voet, 2019).

The results of Model 2 are generally consistent with those reported in Model 1, implying that previous performance achievement may also have a positive impact on budget allocations when fiscally stressed. Such impact, however, could be relatively less substantial considering the smaller magnitude and marginal significance of the coefficient of the interaction term (see also the marginal significance of the likelihood test result). In Model 3, the key variables of interest fail to achieve statistical significance. The allocative efficiency of PBB may recede or disappear if there is a relatively long-time lag between performance achievement and budget allocations. Timely information may be crucial in linking performance achievement to budgetary decisions.

The results of the control variables are generally consistent across the models. As expected, the program budget amount variable significantly impacts program budget changes. Its negative direction indicates that more extensive programs will likely experience smaller or negative budget changes. As expected, the two department-level variables are statistically significant but have different directions. The total number of programs variable has a negative impact on program budget changes, while the opposite is the case for the department budget amount variable. Median household income and unemployment rate are negatively associated with program budget changes. The election-year variable is significant only in Model 1.

Discussion

The above findings suggest the reward-sanction mechanism involved in performance-based resource allocations during fiscally turbulent times. Conversely, program performance achievement may not significantly impact budget allocations when coupled with normal or upturn years. To further explore these findings, Figure 1 depicts these interactive

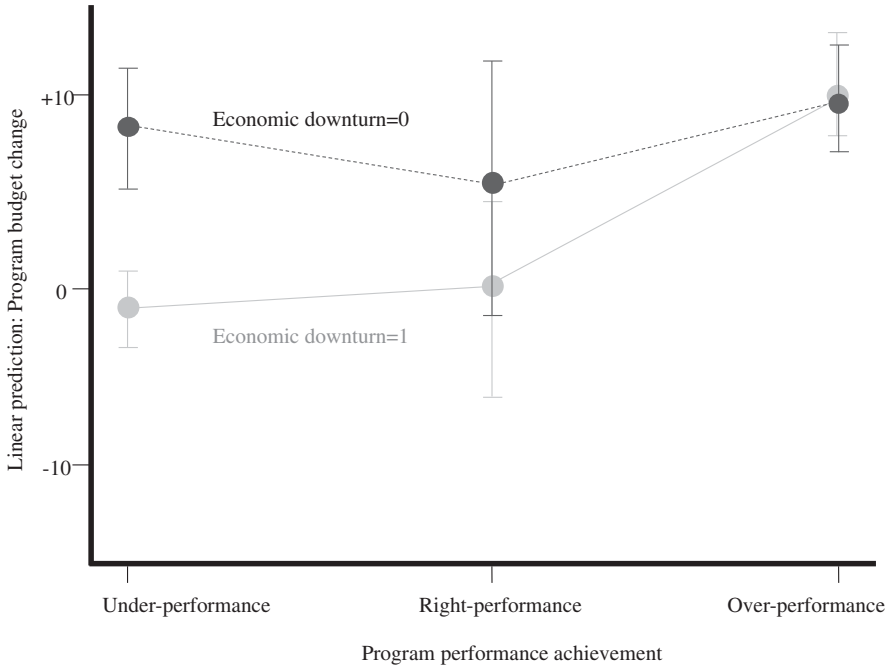


Figure 1. Interaction results.

relationships. The large gap under-performing programs have between fiscally stressed and affluent times disappears when looking at right or over-performing programs. However, the budget-increasing effect of performance achievement is more apparent with programs' over-performance than right-performance. According to Nielson and Baekgaard's (2015) blame-avoidance theory, public officials are motivated to express their support for high-performing programs as it can be a "credit-claiming strategy" (p. 550) for their political gains. As discussed previously, an average level of performance could be a reason for budget cuts because officials must find a way to meet their budgetary constraints—the findings of this study support this idea.

The major findings of this study are directly comparable with some prior studies such as Gilmour and Lewis (2006a, 2006b), Ho (2011), and Ryu (2013) due to a similar focus on program-level information. This study offers evidence consistent with Gilmour and Lewis (2006a, 2006b) and Ho (2011), but stands against Ryu's (2013) null finding of the relationship between program performance and budget allocations. Except for these directly comparable cases, this study should be compared with other agency or department-level studies with caution. For example, Hou et al. (2011) demonstrated no significant allocation effect of PBB at the department or division level during fiscally hard times. The results reported in this study do not necessarily run counter to the authors'

findings. Instead, this study adds to the extant department or agency-level findings by showing that performance-based allocations may still work at the program level, while they may be relatively less functional in explaining department-level results.

Almost 60% of the sample departments included in the analysis process experienced both budgetary rewards and sanctions in downturn years. For example, 17 state executive departments had those programs that experienced budget cuts in 2011 after recording under-performance in 2009. Of the 17 departments, 10 (58.8%) also experienced the reward mechanism, indicating that some of their programs were over-performed and received more budget allocations. This observation does not guarantee direct resource exchanges among each department's programs. It suggests, however, that performance achievement may change resource allocation patterns within a department. Taking the Department of Social and Health, for example, Table 3 shows that with some exceptions of smaller programs, their major programs experienced both the reward and sanction mechanisms in 2011. Note that this is the case for 60% of the sample departments. The remaining 40% may still support the possibility of performance-related resource allocations at the department level.

Table 3. Department of Social and Health Services Example (2011).

	Under-performance	Over-performance
Budget increase or no change	• Preventative services for juveniles • Vocational rehabilitation projects and grants	• Adoption services and support • Disproportionate share hospital/proshare • Family support program for developmentally disabled clients • Medicaid program for aged, blind and disabled • Professional services • Residential substance use disorder treatment services • Special programs
	An average of 5.23 increase Average program budget: $4.9 M	An average of 7.01 increase Average program budget: $73.4 M
Budget decrease	• Administrative costs • Behavioral rehabilitative services • Community services for locally committed juveniles • Family foster home care • Family reconciliation services • Juvenile rehabilitation administration • Nursing home services • Other foster care • Parole transitional services for state committed juvenile offenders	• Alternate response system • Crisis residential center • Financial services administration • Food stamp administration • Optional health benefits: dental, vision, and hearing
	An average of 7.94 decrease Average program budget: $86.6 M	An average of 14.36 decrease Average program budget: $18.0 M

Conclusion

This study aimed to answer three questions: (1) Does performance achievement shape budget allocations? (2) When does performance achievement affect budget allocations? (3) How does the performance-allocation link look? As Robinson and Brumby (2005) pointed out, these questions are critical, but few studies have directly and empirically examined them. This study proposed a set of competing hypotheses from the organizational rationality perspective. The hypotheses were tested based on the PBB experience of Washington State's executive programs from 2006 to 2015. The findings denoted that the performance-based allocation idea may work in certain circumstances. Specifically, this study found the reward-sanction mechanism involved in allocating government resources during a period of fiscal stress. According to the results, the performance-allocation link may not be viable in normal or fiscally affluent times. Fiscal condition may affect public officials' motivation to link performance to resource allocations.

This study has some limitations. First, it focused only on Washington State's public programs. Though Washington offered an appropriate setting for this study, there should be some caveats in generalizing the research findings. The state's unique contexts, such as a long history with PBB and a strong emphasis on objective benchmarks, must be considered appropriately. The limitation could also lead to methodological concerns about potential omitted variable bias and endogeneity, despite this study's significant efforts to resolve such concerns in several ways. Second, measuring government performance is challenging; there is no one correct way of measuring it. Despite some precedents, the measurement used in this study (i.e., under, right, and over-performance) is also imperfect. Third, while previous studies have highly stressed the importance of the qualitative approach to performance information use and PBB (Hou et al., 2011; Zaltsman, 2009), this study limited itself to quantitative analysis. All these reservations warrant future research.

Despite these limitations, this study contributes to the literature. Normatively, performance information should be connected to budgeting (Moynihan, 2003; Robinson, 2007). However, how to link performance to budget allocations is complicated in reality. This complexity has made scholars doubtful about the feasibility of performance-based allocations (e.g., J. E. Melkers et al., 2002). This study does not replace or stand against such skepticism. Instead, as one of the few studies that support the allocative efficiency of PBB, it suggests the ongoing need to revisit this important issue in different contexts. Notably, this study documented the reward-sanction mechanism involved in the performance-allocation link during fiscally hard times. This endeavor significantly advances the literature, given that few studies have directly examined the link.

Under certain conditions, performance achievement still matters in allocating budgetary resources. This conclusion suggests important practical implications: performance information can help inform public officials about their programs and resource efficiency, supporting their challenging decisions during fiscally hard times. This direct link between performance information and budget allocations could add to performance management systems' tangible and intangible benefits, such as communication effects. Doing so may lay consistent and greater emphasis on the need for performance measurement and information in the public sector. Whether such informing mechanisms result in optimal budget decisions or not goes beyond the scope of this study. It is, however, clear from this study's findings that efforts to develop, maintain, and refine government performance management systems should be continued.

Notes

1. Lu's (2011) definition of PBB also supports this idea: "performance budgeting is about bringing the performance information to the same table where resource allocations decisions are made" (p. 7 9). Robinson (2007) stayed on the same line defining PBB as "public sector funding mechanisms and processes designed to strengthen the linkage between funding and results, through the systematic use of formal performance information, with the objective of improving the allocative and technical efficiency of public expenditure" (pp. 4–5).
2. Using the term organizational rationality, the intent is to discuss public officials' motivations, decision and actions at the collective level. Different public officials such as elected officials, executive employees, and budget officers can have different motivations at different budgeting stages with regard to performance information use. In this study, however, the focus is not on a specific group of actors at a specific stage, but on offering a general discussion of their collective-level motivations, decisions and actions.
3. This is an arbitrary measure. As an effort to mitigate this concern, the models were estimated twice, with and without these multi-indicator cases. The results were generally consistent (see Appendix A). To maintain a sufficient number of observations and avoid potential selection bias concerns, this study reports the models including the multi-indicator cases.
4. This study also considered additional socio-economic variables such as state population, age and ethnicity. However, those additional variables were excluded due to collinearity. Some of the control variables were also measured as percentage or percentage point change and entered the models (total department budget amount, median household income, and unemployment rate). The results were consistent, but they resulted in lower R^2 values. Hence, this study decided not to use percentage or percentage point change measures.

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No potential conflict of interest was reported by the author(s).

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Data availability statement

The author confirms that the data supporting the findings of this study are available within the article.

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Appendix A

Table A1. Supplemental Regression Results.

	Dependent variable: annual program budget change (two-year lag)			
Variable	Without multi-indicator cases	Heckman selection second-stage model	Cross-sectional and year fixed-effect model	System GMM estimation
Program performance achievement	0.462	0.030	0.300	−2.446
	(2.169)	(2.314)	(2.188)	(2.710)
Economic downturn	−9.284***	−10.376***	−0.901	−6.943*
	(2.924)	(2.935)	(4.646)	(3.834)
Interaction term (program performance achievement*economic downturn)	5.651*	5.339*	6.045**	8.573***
	(2.882)	(2.986)	(2.889)	(2.961)
Number of performance indicators	–	1.998	1.712	5.410
		(1.288)	(1.399)	(3.439)
Program budget amount (Ln)	−6.055***	−6.137***	−7.273***	−26.742*
	(1.282)	(1.321)	(1.545)	(13.632)
General fund	−4.440	−5.277	−3.134	5.534
	(2.964)	(3.242)	(4.880)	(6.365)
Total number of programs	−0.108*	−0.123**	–	−1.100**
	(0.056)	(0.058)		(0.533)
Department budget amount (Ln)	4.558***	4.559***	–	23.415***
	(0.978)	(1.306)		(8.696)
Median household income	−1.515***	−1.525***	–	−2.955***
	(0.476)	(0.546)		(0.776)
Unemployment rate	−2.867***	−3.033***	–	−4.740***
	(0.707)	(0.757)		(1.277)
Election year	11.652***	11.385***	–	6.290**
	(3.907)	(4.002)		(2.433)
Constant	112.225***	112.843***	–	172.148***
	(30.979)	(35.921)		(57.351)
Lambda	–	2.498	–	–
		(4.002)		
N	1659	1749	1749	1749
F	5.99***	5.12***	4.60***	3.49***
R^2	0.062	0.062	0.108	–
Highest VIF	3.37	4.65	9.04	3.39
First-stage model information	–	Department and year fixed effects included	–	–
Number of instruments	–	–	–	22 (the lags are restricted to two)
Arellano–Bond test (p value)	–	–	–	.996
Sargan and Hansen tests (p value)	–	–	–	>.594

Notes: Unstandardized coefficients are reported; robust standard errors are in parentheses.
*$p < .10$; **$p < .05$; ***$p < .01$.